

MRI Image Quality Metrics (IQMs)

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Abstract

This document summarizes commonly used MRI image quality metrics (IQMs) and their mathematical definitions for reproducibility and reporting.

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Introduction

This document summarizes commonly used MRI image quality metrics (IQMs) and their mathematical definitions for reproducibility and reporting.

1 Signal-to-Noise Ratio (SNR)

$$\text{SNR} = \frac{\mu_{\text{signal}}}{\sigma_{\text{noise}}} \quad (1)$$

where μ_{signal} is the mean intensity in the signal region and σ_{noise} is the standard deviation in a background noise region.

[1]

References

- [1] Michael Elad. *Sparse and Redundant Representations: From Theory to Applications in Signal and Image Processing*. Springer, 2010.